IT SERVICE DESK TICKET ANALYSIS USING PYTHON

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FINAL PROJECT SUMMARY

SUBMITTED BY

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1. PROJECT OVERVIEW

This project analyzes IT Service Desk Ticket data using Python to gain insights into IT support operations. The dataset was cleaned, transformed, and analyzed using Google Colab. Exploratory Data Analysis (EDA), statistical analysis, and visualizations were performed to identify ticket trends and support performance. The project also examined ticket priorities, resolution status, SLA breaches, and agent workload. These insights help improve IT service efficiency and support better decision-making. Overall, the project demonstrates the use of data analytics to optimize IT service management.

2. DATA SOURCE

Dataset Name	IT Service Desk Tickets Dataset (Main Table: tickets.csv)
Source	Hugging Face — Mindweave Help Desk Tickets Dataset
File Type	CSV
Main Dataset	tickets.csv
Number of Records	1,000
Number of Attributes	15

The primary dataset used in this project is tickets.csv, which contains IT service desk ticket information such as ticket ID, creation date, priority, status, support channel, assigned agent, requester department, affected service, and resolution details. Additional datasets (agents.csv, categories.csv, and sla_breaches.csv) were merged with the main dataset to enrich the analysis, while comments.csv was excluded because it was not relevant to the project objectives.

3. PROBLEM STATEMENT

IT service desk teams receive a large number of support tickets every day, making it difficult to monitor ticket trends, prioritize requests, and ensure timely resolution. Without proper data analysis, organizations may face delays in resolving issues, increased SLA breaches, uneven workload distribution among support agents, and reduced service quality. This project aims to analyze IT service desk ticket data using Python to identify patterns, evaluate support performance, and generate insights that help improve operational efficiency and support better decision-making.

4. OBJECTIVES

- Analyze the overall distribution and monthly trends of support tickets.
- Evaluate ticket resolution performance across priority levels.
- Identify common ticket categories, affected services, and support channels.

- Assess support agent workload distribution.
- Examine SLA breaches, escalations, and outages.
- Provide recommendations to improve IT service desk performance.

5. DATASET ATTRIBUTES

Attribute	Description
ticket_id	Unique ticket identifier
created_at	Ticket creation date and time
first_response_at	First response date and time
resolved_at	Ticket resolution date and time
priority	Ticket priority
status	Ticket status
channel	Support channel
requester_department	Requester department
category_id	Category identifier
assigned_agent_id	Assigned agent identifier
actual_hours	Resolution time (hours)
escalated	Escalation flag
outage_related	Outage indicator
summary	Brief issue summary
description	Detailed issue description

6. TOOLS & TECHNOLOGIES

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn

7. DATA PREPROCESSING & FEATURE ENGINEERING

The dataset was prepared for analysis using various data preprocessing and feature engineering techniques to improve data quality and ensure accurate analysis.

Data Cleaning Steps

- Checked the data types of all columns.
- Identified missing values in the dataset.
- Retained missing values in the resolved_at column, as they represent unresolved tickets.
- Checked for duplicate records and confirmed that no duplicates were present.
- Removed unnecessary columns (summary and description) from the analysis.
- Converted the required columns to appropriate data types (DateTime and Category).
- Merged the required datasets (tickets, categories, agents, and sla_breaches) using left joins.

Feature Engineering

New features were created to enhance the analysis and support the project objectives.

New Features Created:

- Year was extracted from the created_at column.
- Month was extracted from the created_at column.
- Day was extracted from the created_at column.
- Quarter was extracted from the created_at column.
- Resolution_Status was created to identify whether a ticket was resolved or unresolved.
- SLA_Status was created to classify tickets as SLA Met or SLA Breached.

8. ANALYSIS AND VISUALIZATIONS

The dataset was analyzed using Exploratory Data Analysis (EDA) and statistical visualization techniques to identify ticket trends, support performance, SLA compliance, agent workload, and relationships between different variables. Various univariate, bivariate, and multivariate analyses were performed to generate meaningful insights from the IT service desk data.

9. EXPLORATORY DATA ANALYSIS (EDA)

Initial inspection of the dataset included:

- Data frame structure verification using head(), info(), and shape.
- Statistical summary of numerical columns using describe().
- Missing value identification and verification.
- Duplicate record validation.
- Data type verification of all columns.

EDA helped in understanding the structure, quality, and overall characteristics of the IT service desk ticket data.

10. STATISTICAL ANALYSIS

Statistical measures were calculated for the **actual_hours** and **breach_minutes** columns to summarize their distribution and variability.

Measures included:

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Skewness
- Kurtosis

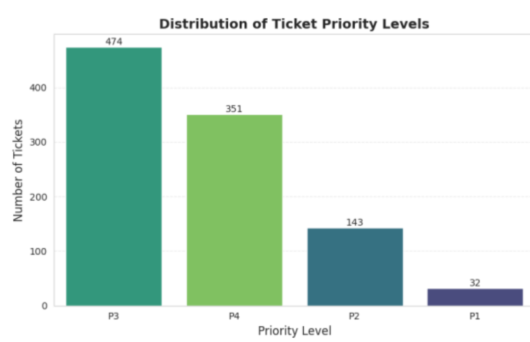
These measures helped in understanding the central tendency, variability, and distribution of the ticket resolution time (**actual_hours**) and SLA breach duration (**breach_minutes**) data.

11. DATA VISUALIZATION SUMMARY

11.1 Univariate Analysis

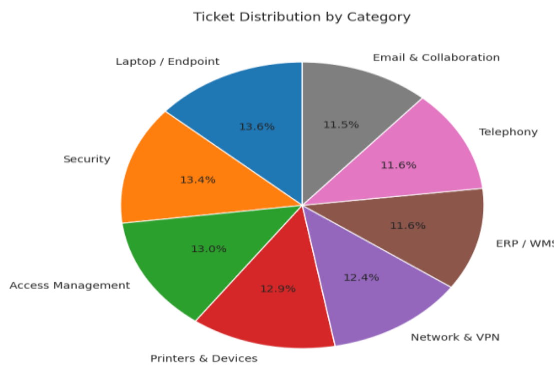
Univariate analysis was performed to analyze one variable at a time.

1. Distribution of Ticket Priority (Count Plot)



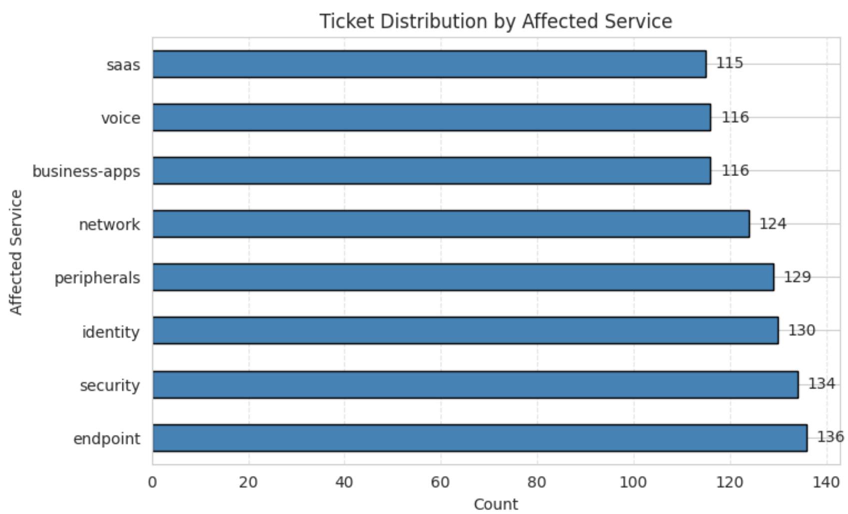
> **Interpretation:** Most tickets were assigned **Medium** priority, followed by **Low** and **High** priorities.

2. Distribution of Ticket Category (Pie Chart)



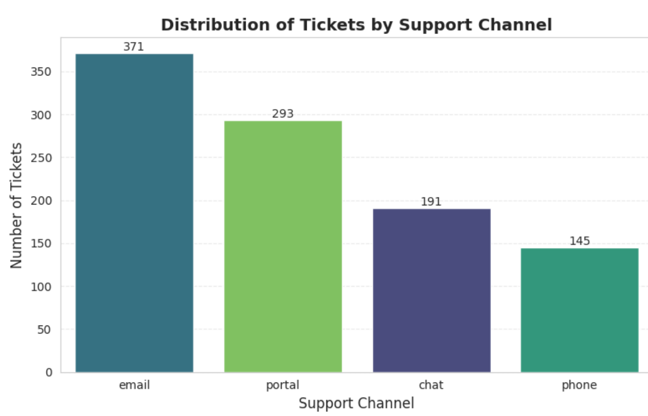
Interpretation: The **Laptop/Endpoint** category recorded the highest number of support tickets.

3. Distribution of Affected Service (Horizontal Bar Chart)



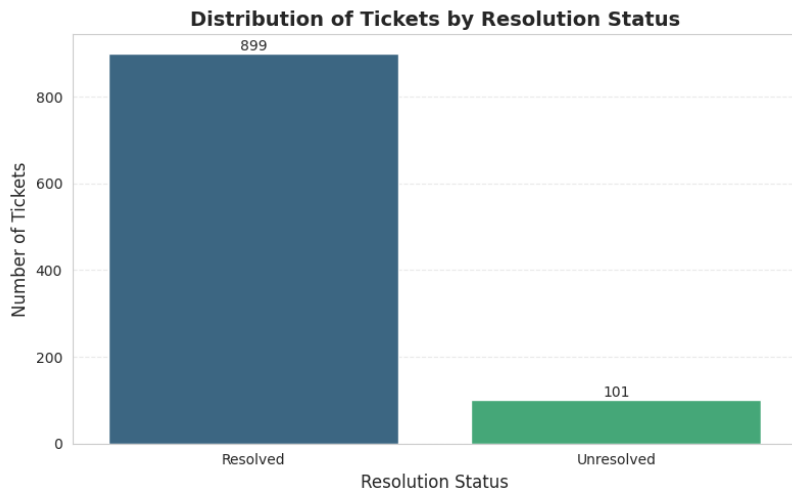
Interpretation: Endpoint was the most frequently affected IT service.

4. Distribution of Support Channel (Count Plot)



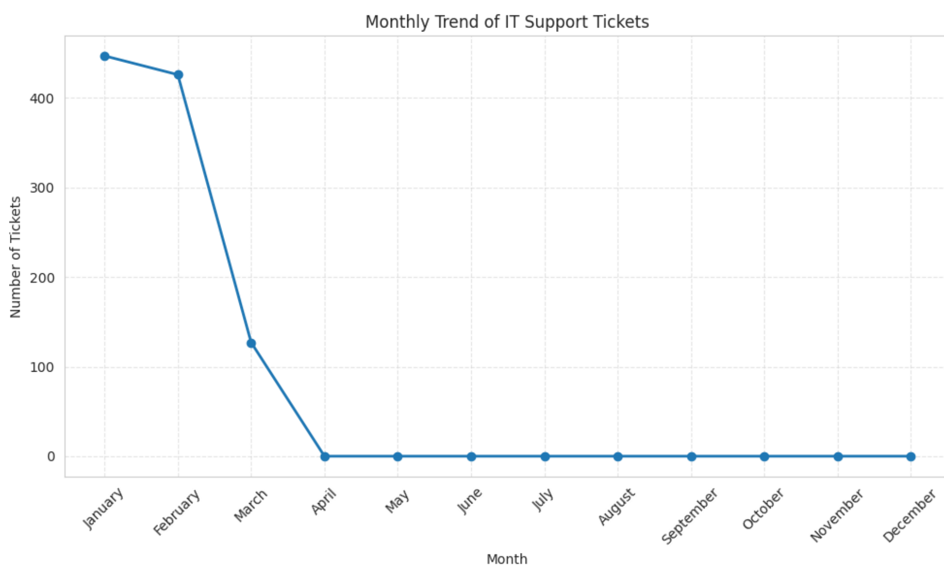
Interpretation: Email was the most commonly used support channel for raising tickets.

5. Distribution of Resolution Status (Count Plot)



> **Interpretation:** The majority of tickets were successfully **resolved (899)**, while a smaller number remained **unresolved (101)**.

6. Monthly Ticket Trend (Line Chart)

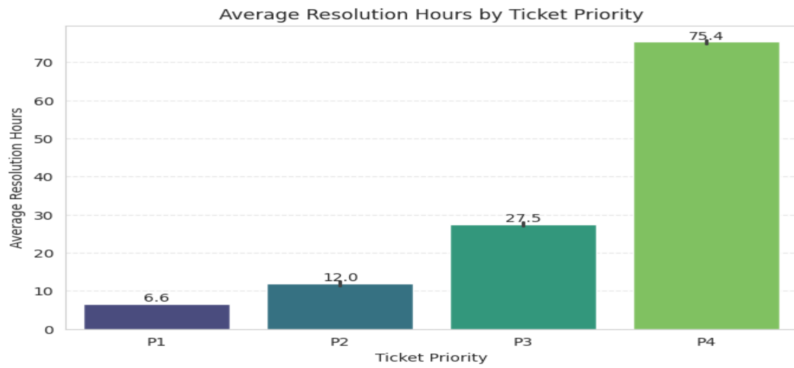


> **Interpretation: January** recorded the highest number of tickets (**447**), followed by February, while March had the lowest ticket count.

11.2 Bivariate Analysis

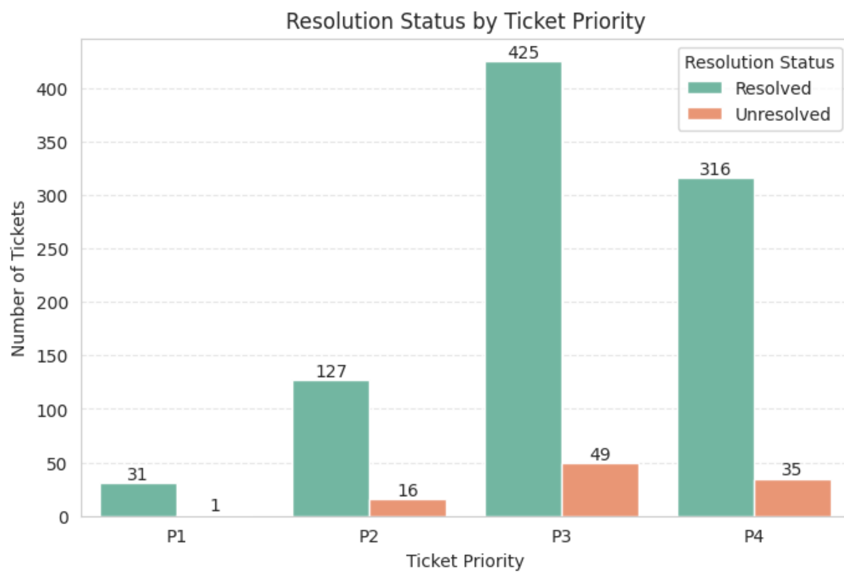
Bivariate analysis was performed to study relationships between two variables.

1. Ticket Priority vs. Average Resolution Time (Bar Plot)



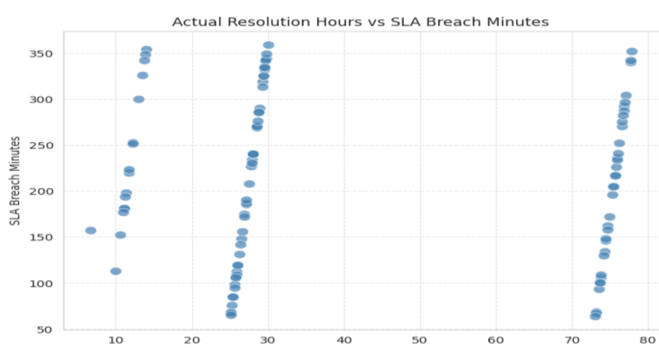
Interpretation: Resolution time varied across ticket priority levels, ranging from **P1 (6.6 hours)** to **P4 (75.4 hours)**.

2. Ticket Priority vs. Resolution Status (Grouped Bar Chart)



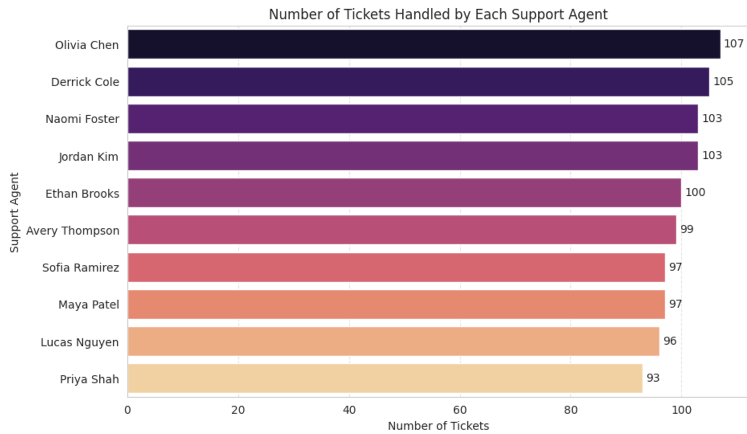
Interpretation: Across all priority levels, **resolved** tickets outnumbered unresolved tickets. **P3** recorded the highest number of resolved tickets (**425**).

3. Actual Hours vs. Breach Minutes (Scatter Plot)



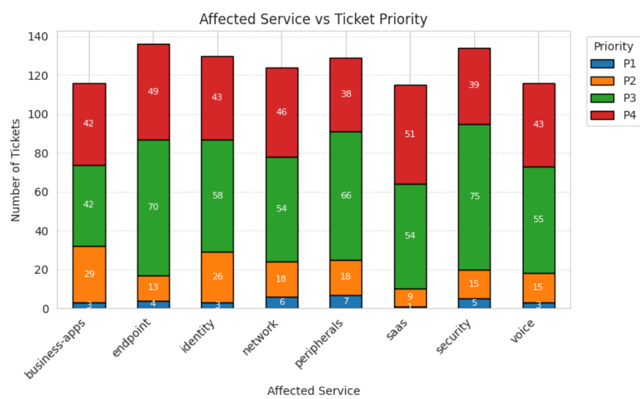
Interpretation: Tickets with **higher resolution hours** generally showed **higher SLA breach durations**.

4. Assigned Agent vs. Number of Tickets (Horizontal Bar Chart)



> **Interpretation: Olivia Chen** handled the highest number of tickets (**107**), while **Priya Shah** handled the fewest (**93**).

5. Affected Service vs. Ticket Priority (Stacked Bar Chart)

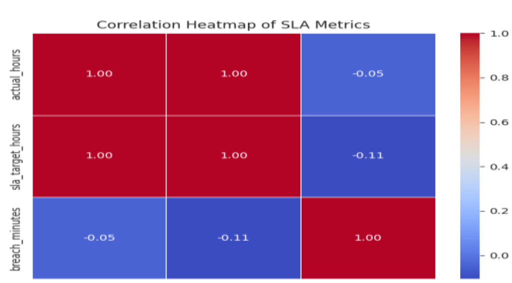


> **Interpretation:** Most affected services are dominated by **P3** and **P4** tickets. Services such as **Security**, **Endpoint**, and **Peripherals** handle a relatively higher volume of tickets.

11.3 Multivariate Analysis

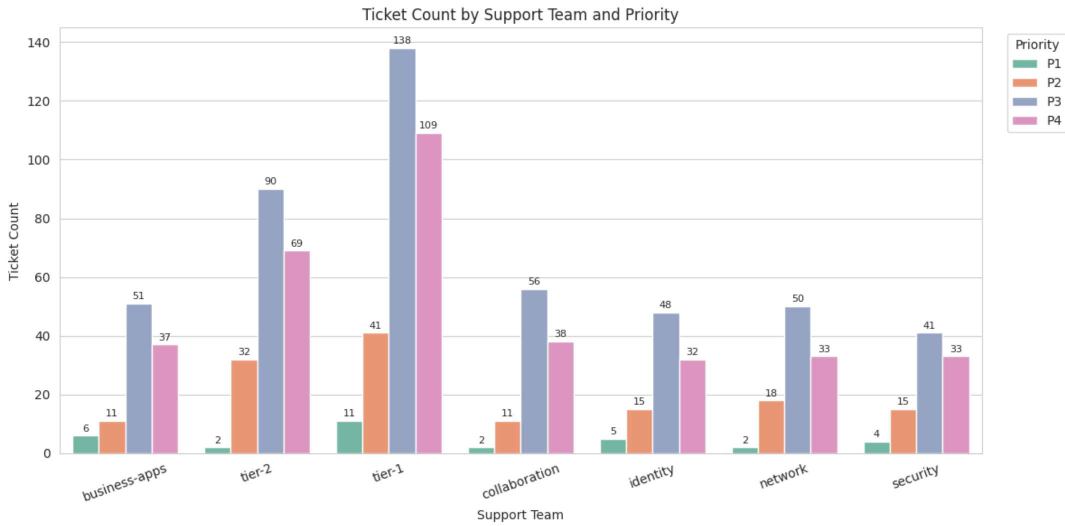
Multivariate analysis was performed to analyze relationships among multiple variables.

1. **Correlation Analysis (Correlation Heatmap)**



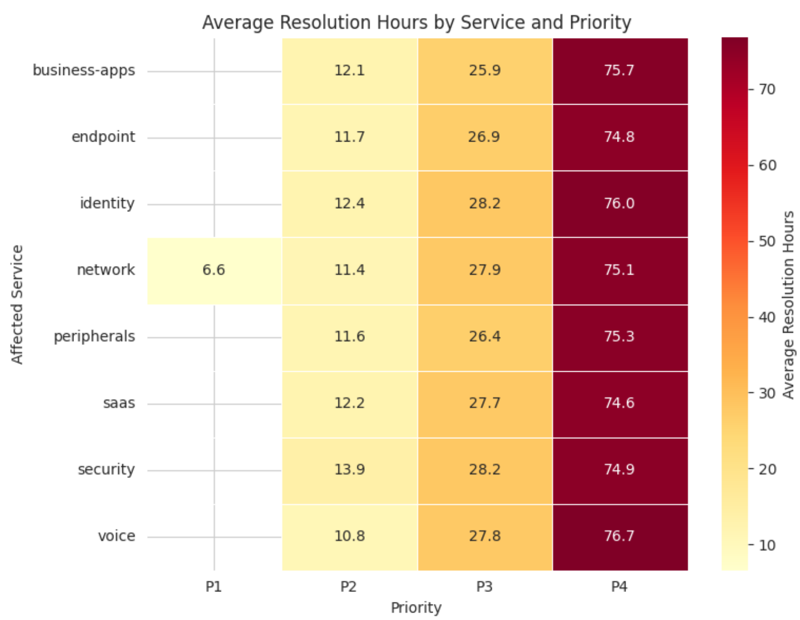
> **Interpretation: Actual Resolution Hours** and **SLA Target Hours** have a very strong positive correlation (**1.00**), indicating they increase together.

2. Support Team vs. Ticket Priority (Grouped Bar Chart)



> **Interpretation: Tier-1** handled the highest number of tickets across all priority levels, especially **P3 (138)** and **P4 (109)**.

3. Priority, Resolution Time, and SLA Breach Analysis (Pivot Table Heatmap)



> **Interpretation: P4** tickets have the highest average resolution hours across all services. **Voice (76.7 hrs)** and **Identity (76.0 hrs)** require the longest resolution time for P4 tickets.

12. SUMMARY OF FINDINGS

1. **January (447)** and **February (426)** recorded the highest number of support tickets, while **March (127)** had the lowest ticket volume.
2. **P3 (474)** and **P4 (351)** received the highest number of support tickets.
3. **899** out of 1,000 tickets were **resolved**, while **101** remained **unresolved**.
4. **Laptop/Endpoint (138)** was the most common ticket category, **Endpoint (136)** was the most affected service, and **Email (371)** was the most used support channel.
5. **Tier-1** handled the highest workload, with **138 P3 tickets** and **109 P4 tickets**.
6. **101 tickets** breached the SLA, **116 tickets** were escalated, and no outage-related tickets were reported, indicating areas for service improvement.
7. Focus on reducing SLA breaches (101), minimizing escalated tickets (116), and balancing Tier-1 workload to improve overall service desk performance.

13. TYPES OF ANALYSIS

13.1 Descriptive Analysis

Descriptive analysis summarizes historical IT service desk ticket data.

Examples:

- Ticket priority distribution
- Monthly ticket trend
- Resolution status distribution
- Support channel and category distribution

13.2 Diagnostic Analysis

Diagnostic analysis explains the reasons behind ticket patterns and service performance.

Examples:

- Relationship between ticket priority and resolution time
- Impact of SLA breaches on ticket resolution
- Workload distribution among support agents

13.3 Predictive Analysis

Predictive analysis helps forecast future service desk performance based on historical data.

Examples:

- Predicting future ticket volume
- Forecasting ticket resolution time
- Identifying potential SLA breach trends

13.4 Prescriptive Analysis

Prescriptive analysis suggests actions to improve IT service desk operations based on analytical insights.

Examples:

- Optimize support agent workload distribution
- Prioritize high-priority tickets for faster resolution
- Implement measures to reduce SLA breaches and improve service efficiency

14. FUTURE ENHANCEMENTS AND SUGGESTIONS

- Include a larger and more recent IT service desk dataset for improved analysis.
- Apply machine learning models to predict ticket resolution time and SLA breaches.
- Develop an interactive dashboard for real-time monitoring of IT support performance.
- Automate the data analysis and reporting process to generate regular performance reports.

15. CONCLUSION

This project analyzed IT service desk ticket data using Python to identify trends and support performance. Data preprocessing, statistical analysis, and visualizations provided meaningful operational insights. The analysis highlighted ticket priorities, resolution time, SLA breaches, and agent workload distribution. Overall, the project demonstrates how data analytics can improve IT service management and support decision-making.